

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering ASSESSMENT TOOL 2 – Data Interpretation

|                  |                                                                                                                                            |               |                                         |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------|
| Course Code:     | DSA0405                                                                                                                                    | Course Title: | FUNDAMENTALS OF DATA SCIENCE            |
| CO Assessed:     | CO3 – Data Interpretation                                                                                                                  | Assessment:   | Assessment Tool 2 – Data Interpretation |
| Weightage:       | 40% of CIA                                                                                                                                 | Total Marks:  | 100                                     |
| CO3 Statement:   | <i>Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)</i> |               |                                         |
| Student Name:    |                                                                                                                                            | Reg. No.:     |                                         |
| Section / Batch: |                                                                                                                                            | Date:         |                                         |

### Code of Conduct

I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: \_\_\_\_\_

| Criteria                | Weight age | Level 4 – Exemplary                                              | Level 3 – Proficient                           | Level 2 – Developing                    | Level 1 – Beginning                                  | Marks |
|-------------------------|------------|------------------------------------------------------------------|------------------------------------------------|-----------------------------------------|------------------------------------------------------|-------|
| Understanding of Case   | 25         | Thorough understanding; clearly identifies all key issues        | Good understanding with most issues identified | Basic understanding; some issues missed | Poor understanding; problem not identified correctly |       |
| Application of Concepts | 25         | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps   | Limited application of concepts         | Fails to apply relevant concepts                     |       |
| Analysis                | 20         | In-depth analysis with strong reasoning and insights             | Good analysis with logical reasoning           | Basic analysis with limited depth       | Weak or no analysis; lacks reasoning                 |       |
| Solution Design         | 20         | Innovative, feasible solutions with strong justification         | Logical solutions with adequate justification  | Basic solutions with weak justification | Incorrect or no solution provided                    |       |
| Clarity                 | 10         | Clear, well-structured, and easy to understand                   | Organized and understandable presentation      | Limited clarity and structure           | Poor clarity; difficult to understand                |       |

# Descriptive Statistics – Assignment Answers

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## 11. Standardization / Normalization

### The Problem

The dataset has two features on wildly different scales: “Age” ranges from 18 to 75 (a span of about 57 units), while “Income” ranges from ₹20,000 to ₹20,00,000 (a span of nearly 19,80,000 units). Both are being fed into a distance-based algorithm such as K-Nearest Neighbours, K-Means clustering, or SVM with an RBF kernel, all of which rely on Euclidean (or similar) distance to decide how “close” two data points are.

### Why Failing to Standardize Distorts the Analysis

Distance-based algorithms compute distance between two points as the square root of the sum of squared differences across all features:  $d = \sqrt{(\text{Age}_1 - \text{Age}_2)^2 + (\text{Income}_1 - \text{Income}_2)^2 + \dots}$ . Because Income's raw values are on the order of lakhs while Age's raw values are on the order of tens, even a tiny percentage difference in Income (say, ₹5,000 out of ₹20,00,000, which is only 0.25% of its range) produces a squared term that is astronomically larger than the squared term contributed by a large percentage difference in Age (say, 20 years out of a 57-year range, which is about 35% of its range).

In practical terms, the algorithm effectively becomes an “Income-only” algorithm. Two people who are 18 and 75 years old (completely different life stages) but happen to have similar incomes would be judged as “near neighbours,” while two people who are the same age but differ in income by a few thousand rupees would be judged as “far apart.” This defeats the purpose of using both features — the model is not actually learning from Age at all; Income silently dominates every distance calculation purely because of its numeric magnitude, not because it is more informative.

This is not a cosmetic issue — it changes cluster assignments in K-Means, changes which neighbours are picked in KNN, and changes the support vectors chosen in SVM. Standardization (z-score scaling) or normalization (min-max scaling) removes this artificial dominance by putting every feature on a comparable scale (typically mean 0, SD 1 for standardization, or [0,1] for

normalization), so that distance reflects genuine dissimilarity across all features rather than whichever feature happens to have the largest raw numbers.

### **Z-Score Calculation for Age = 45**

The z-score formula is:

$$z = (x - \mu) / \sigma$$

Given  $x = 45$ ,  $\mu$  (mean) = 40, and  $\sigma$  (SD) = 12:

$$z = (45 - 40) / 12 = 5 / 12 = 0.4167 \text{ (approximately)}$$

A z-score of about 0.42 means that an Age value of 45 lies roughly 0.42 standard deviations above the mean age of 40 — a mild, unremarkable deviation, not an outlier. On the original 18–75 scale this same value looked like just another number, but the z-score expresses it in a common, comparable unit that can be placed directly alongside the standardized Income figures without one variable overpowering the other.

## **12. Distribution Shape from Summary Statistics Alone**

### **What Skewness $\approx 0$ and Equal Mean/Median/Mode Tell Us**

Mean = Median = Mode = 45 together with skewness  $\approx 0$  indicate that the distribution is symmetric — there is no long tail pulling the distribution to the left or right. On the basis of central tendency and skewness alone, one might be tempted to assume this dataset looks like a well-behaved, textbook normal distribution centered at 45.

### **What Kurtosis = 5.8 Adds**

Kurtosis measures the “peakedness” and tail-heaviness of a distribution relative to a normal distribution, which has a kurtosis of 3 (or excess kurtosis of 0). A kurtosis of 5.8 is well above 3, classifying the distribution as leptokurtic. This tells us something that symmetry and central tendency cannot: the distribution has a sharper, taller peak around the mean than a normal curve, combined with fatter tails — meaning extreme values (both very high and very low, in a way that keeps things symmetric) occur more frequently than a normal distribution would predict.

### **Why This Combination Matters**

Symmetry alone (skewness  $\approx 0$ ) only rules out lopsidedness — it says nothing about how concentrated or spread out the data is around the center. Two datasets can both have Mean = Median = Mode = 45 and skewness  $\approx 0$ , yet one could be a gentle, evenly-spread normal curve while the other is a leptokurtic distribution with most values tightly clustered near 45 but occasional sharp outliers far from it. Relying only on mean, median, mode, and skewness would miss this entirely and could lead to underestimating risk or variability — for instance, in a finance context, a leptokurtic “fat-tailed” return distribution implies a higher chance of extreme gains or losses than a normal distribution with the same mean would suggest, even though the average and symmetry look perfectly ordinary. Kurtosis is therefore essential to complete the picture of the distribution's true shape, especially for identifying tail risk.

### 13. Grouped Data — Mean and Variance (Step-Deviation Method)

#### Step 1: Setting Up Midpoints and Assumed Mean

The midpoint of each class interval is the average of its lower and upper limits. We choose an assumed mean (A) at the midpoint of a central class — here,  $A = 25$  (midpoint of the 20–30 class) — and a class width  $h = 10$ , since step-deviation uses  $d = (x - A) / h$  to keep the numbers small and easy to work with.

| Class Interval | Frequency (f) | Midpoint (x) | $d = (x-25)/10$ | $f \times d$ | $f \times d^2$ |
|----------------|---------------|--------------|-----------------|--------------|----------------|
| 0–10           | 5             | 5            | -2              | -10          | 20             |
| 10–20          | 8             | 15           | -1              | -8           | 8              |
| 20–30          | 15            | 25           | 0               | 0            | 0              |
| 30–40          | 12            | 35           | 1               | 12           | 12             |
| 40–50          | 10            | 45           | 2               | 20           | 40             |

Totals:  $N = \Sigma f = 5 + 8 + 15 + 12 + 10 = 50$ ,  $\Sigma fd = -10 - 8 + 0 + 12 + 20 = 14$ ,  $\Sigma fd^2 = 20 + 8 + 0 + 12 + 40 = 80$ .

#### Step 2: Mean by Step-Deviation Method

$$\text{Mean} = A + h \times (\Sigma fd / N)$$

$$\text{Mean} = 25 + 10 \times (14 / 50) = 25 + 2.8 = 27.8$$

#### Step 3: Variance by Step-Deviation Method

$$\text{Variance} = h^2 \times [ (\Sigma fd^2 / N) - (\Sigma fd / N)^2 ]$$

$$\text{Variance} = 100 \times [ (80/50) - (14/50)^2 ] = 100 \times [1.6 - 0.0784] = 100 \times 1.5216 = 152.16$$

So the mean of the grouped distribution is 27.8, and the variance is 152.16 (giving a standard deviation of approximately  $\sqrt{152.16} \approx 12.34$ ). The step-deviation method avoids working directly with large midpoint values by re-centering around the assumed mean and scaling down by the class width, then correcting for both in the final formulas.

## 14. Comparing Two Distributions via Coefficient of Variation

### Why Standard Deviation Alone Is Misleading Here

Portfolio B has a higher standard deviation (5%) than Portfolio A (3%), which on the surface supports the claim that B is “riskier in absolute terms.” However, standard deviation is an absolute measure of spread expressed in the same units as the data (percentage return here), and it does not account for the fact that B also has a much higher mean return (18% vs 12%). A larger SD is expected, and even desirable, when the average return itself is larger — so comparing raw SDs across portfolios with different means can be misleading.

### Coefficient of Variation (CV) Calculation

$$CV = (SD / \text{Mean}) \times 100$$

For Portfolio A:

$$CV_a = (3 / 12) \times 100 = 25\%$$

For Portfolio B:

$$CV^B = (5 / 18) \times 100 \approx 27.78\%$$

### What CV Captures That SD Alone Doesn't

CV expresses risk (SD) relative to the reward (mean return), making it a measure of risk “per unit” of return — essentially, how much volatility an investor must tolerate for each percentage point of expected gain. This allows fair comparison between two investments that operate on completely different scales of return.

### **Evaluating the Investor's Claim**

The claim has two parts, and the CV analysis shows both are essentially correct. First, “riskier in absolute terms” is confirmed: Portfolio B's SD of 5% is indeed larger than Portfolio A's 3% in raw terms. Second, “not necessarily worse per unit of return” is also broadly supported:  $CV_A$  (25%) and  $CV_B$  (27.78%) are fairly close, with B only slightly higher. This means that although B swings more in absolute percentage points, it is not dramatically worse when that risk is weighed against its considerably higher average return — an investor is not paying a hugely disproportionate risk premium for B's extra return. A purely SD-based comparison would have overstated how much riskier B is; CV reveals that B's extra volatility is largely proportionate to (and justified by) its extra return, though it is still marginally less efficient per unit of risk than A.

## **15. Outlier's Effect on Variance**

### **The Setup**

A dataset of  $n = 10$  values originally has variance = 25 and a mean of 42. One value that was genuinely close to the mean (40) has been mistakenly recorded as 400 — a data-entry error that inflates a typical, unremarkable observation into an extreme outlier.

### **(a) Effect on the Mean**

The mean will increase, but only moderately, because the mean is an average and the error is diluted across all  $n = 10$  values. The error introduced is  $(400 - 40) = 360$ , and since the mean shifts by the error divided by  $n$ , the mean rises by approximately  $360 / 10 = 36$ , pushing it from about 42 to roughly 78. This is a real and noticeable shift, but it is dampened — one bad value among ten cannot single-handedly relocate the mean by the full size of the error; its impact is “spread out” across the whole dataset.

### **(b) Effect on the Variance**

The variance will increase far more dramatically — disproportionately more than the mean. Even a rough, directional estimate makes this clear: the erroneous point's deviation from the (new) mean will itself be large (roughly  $400 - 78 \approx 322$ ), and variance is built from the square of such deviations. Squaring a deviation of this size produces a contribution in the tens of thousands ( $322^2$

$\approx 1,03,684$ ) divided among 10 points — dwarfing the original variance of 25 by several orders of magnitude. The variance will likely balloon into the thousands, not merely double or triple.

### **Why Variance Is Affected Disproportionately More**

The core reason is that variance is calculated using squared deviations from the mean, while the mean itself is a simple linear average. A linear statistic (the mean) responds to an error in proportion to the size of that error divided by  $n$  — a one-time, dampened effect. A quadratic statistic (the variance) responds to the square of the deviation, so a single large error does not just shift the statistic — it explodes it. Doubling the size of an outlier's deviation from the mean roughly quadruples its contribution to the variance, whereas it only doubles its contribution to the mean. This is precisely why variance (and SD) are described as highly sensitive to outliers, and why a single data-entry mistake like recording 40 as 400 can make a dataset's spread look wildly different from its true, underlying variability, even though only one of ten values was affected and the mean itself moved by a comparatively modest amount.